MIS772 – Predictive Analytics

T2 2024

Assignment 2 – Individual
Student name: Phuong Viet Dang
Student number: 224383373

Executive summary (1 page)

1.1. Executive Problem Statement

The task at hand is to estimate the review score ratings of properties listed on a rental platform. For property managers and hosts, maintaining high review scores is critical for driving bookings, improving occupancy rates, and maintaining a competitive advantage. This predictive model aims to help property managers understand what factors influence review scores, enabling them to make informed decisions about optimizing property attributes, service levels, and pricing to meet customer expectations.

The high-level objective is to develop an analytical solution to estimate a property's review score based on predictor attributes in the dataset. These attributes include factors such as property location, number of reviews, amenities offered, and customer sentiment. The solution will provide a data-driven approach to predict the review score of properties, offering insight into which factors contribute most to customer satisfaction.

1.2. Executive Solution Statement

The analysis used a predictive model to estimate review scores, addressing the need to optimize property attributes to enhance customer satisfaction. The model incorporated several predictive techniques, ultimately favoring a Voting Ensemble method, which combined Random Forest, Gradient Boosting, and Decision Trees. This approach provided more accurate predictions than using any individual model alone.

Our recommendation is for property managers and hosts to focus on optimizing key predictor attributes identified by the model, such as location, pricing, and amenities, which have been shown to significantly impact review scores. By making targeted adjustments based on these predictions, managers can expect to improve their properties' review ratings, driving higher bookings and customer satisfaction.

The anticipated benefits of this solution include improved occupancy rates and revenue, enhanced customer satisfaction, and a better understanding of which property features and services are most valued by guests.

Recommendation: Airbnb should encourage hosts to continue using positive language, but also focus on improving property quality and customer experience, as reviews show that sentiment is affected by more than just description tone.

Anticipated Benefits: Aligning host descriptions with customer experiences can improve transparency and trust. Providing guidance to hosts on writing effective descriptions may improve customer satisfaction and positively impact future bookings.

2.2. Data exploration and Attribute selection

The first step in analysis was to explore the datasets (Listing_va.xlsx and reviews_va.xlsx) to select relevant attributes and prepare the data for further processing. In this case, we are primarily concerned with the text attributes that describe the property and review from customers.

Listing dataset attributes using for task A: ID, Description

Review dataset attributes using for task A: Listing ID (match with ID in listing dataset), Comment

Handling missing values: The number of missing in description and comment take a small proportion of the whole dataset, therefore, a filter example operator was used to remove those missing values.

For task B, a subset of attributes has been used to building a model for predating future review score:

1. Host_since: experienced hosts may provide better services, leading to higher review score.
2. Host_total_listing_count: With more properties, host may have refined their process for managing which will impact the quality of services and affect review score.
3. Latitude and longtitude: Location is one of the most important roles in customer satisfaction.
4. Neighbourhood_group_cleansed: The neighbourhood of the property also play a significant impact to customer needs; safer and fun neighbourhood may lead to a better experience.
5. Price: affects the perceived value for money, making it a strong predictor of whether a guest feels satisfied with their stay, impacting the review score.
6. Property_type: reflects the type of experience the guest expects based on the nature of the accommodation, which can influence their satisfaction and, consequently, their review score.
7. Room_type: impacts the level of privacy and comfort, key elements of guest satisfaction, which are critical for determining review scores.
8. Review_score_rating: This is the label attribute since the predictive model is focus on review score

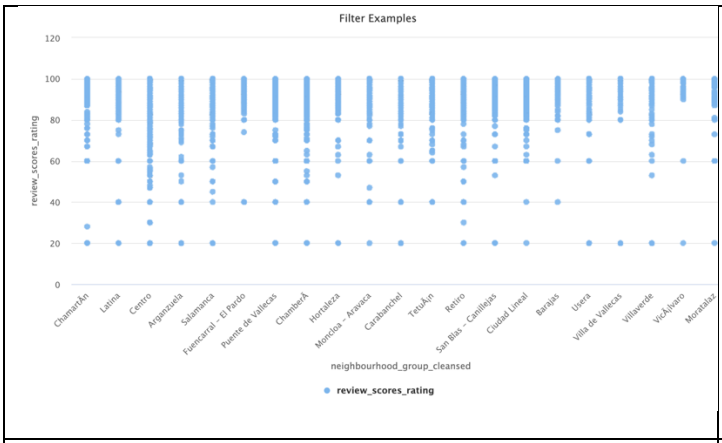


Figure 1.1. Relationship between Review score and neightbourhood

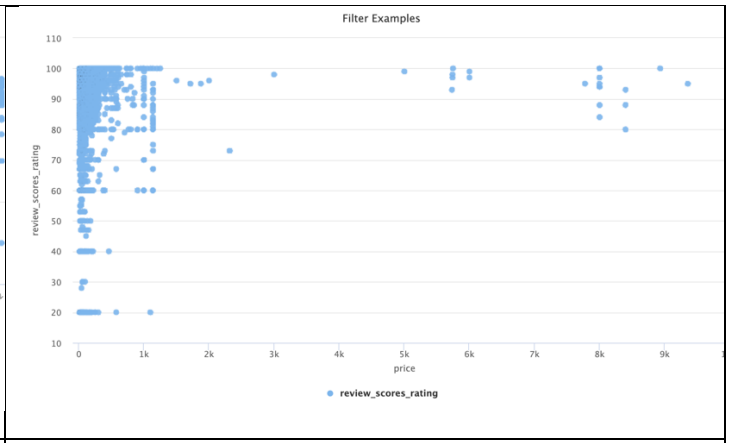


Figure 1.2. Relationship between Review score and Price

Neighbourhood Quality and Review Scores: Central neighborhoods like *Centro*, *Salamanca*, and *Chamberí* consistently receive high review scores, likely due to their central location, amenities, or property quality. Areas like *Puente de Vallecas*, *Villa de Vallecas*, and *Usera* show greater variation, suggesting inconsistent property quality and mixed guest experiences.

Cluster of Low-Priced Listings: Properties priced below 1,000 tend to have high review scores (80-100). However, some outliers exist with lower ratings (20-40), potentially due to property or service issues.

Higher-Priced Listings: Properties priced above 1,000 shows less variation in review scores, with most maintaining high ratings (80-100). There are fewer high-priced listings, but they generally perform well.

Outliers: Some lower-priced properties (<1,000) have very low ratings, pointing to issues beyond price, such as property condition or location.

There is a noticeable lack of low-rated properties in the higher price ranges.

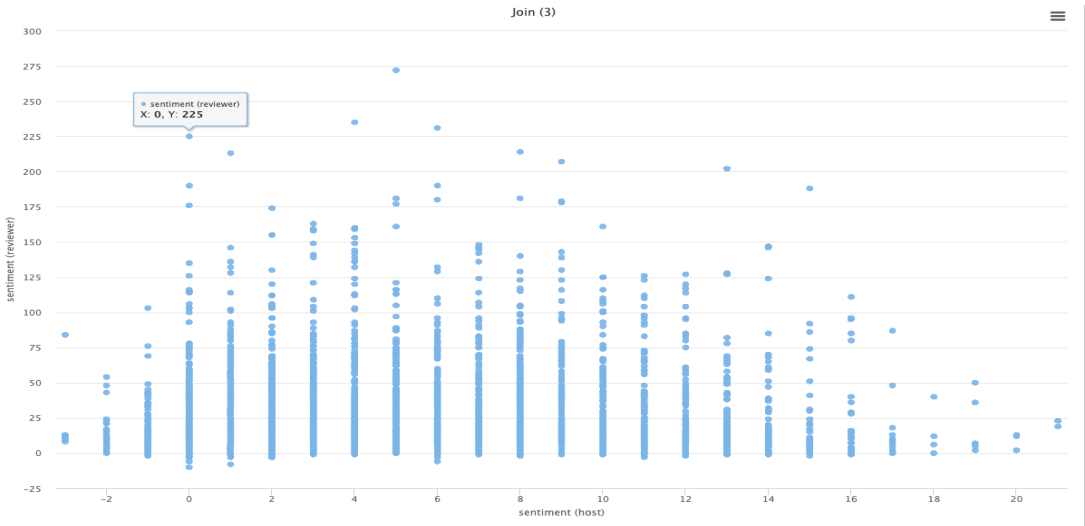
Data Transformation: Several transformations were applied to the dataset to prepare it for model development:

Nominal to Numerical Conversion: Categorical variables, such as property type, were converted into numerical values using the **Nominal to Numerical** operator. This transformation allowed the algorithms to interpret these variables in a meaningful way.

Normalization: The **Normalize** operator was applied to standardize the ranges of the numerical features. This step was essential for ensuring that no single attribute dominated the model simply due to its scale.

2.2. Text mining and Sentiment Analysis:

Attributes	sum(p...	sum(ne...	sentiment (host)	sum(p...	sum(ne...	sentiment (reviewer) ↓
sentiment (reviewer)	0.192	0.179	0.165	0.990	0.812	1
sum(poscount_r)	0.200	0.187	0.172	1	0.887	0.990
sum(negcount_r)	0.198	0.185	0.170	0.887	1	0.812
sum(poscount_h)	1	0.768	0.943	0.200	0.198	0.192
sum(negcount_h)	0.768	1	0.511	0.187	0.185	0.1980872568838809
sentiment (host)	0.943	0.511	1	0.172	0.170	0.165



2.3. Correlation Matrix Interpretation:

The correlation matrix shows the relationships between sentiment attributes. Key findings include:

Sentiment (reviewer) vs. Sentiment (host): The correlation is **0.165**, indicating a **weak positive correlation**. While there is a slight link between host sentiment and review sentiment, it's not strong. Other factors likely play a significant role in shaping reviewer sentiment.

Positive and Negative Sentiment Counts: The sum of positive sentiments shows stronger internal correlations, suggesting that positive interactions are more likely to be consistent across host and reviewer sentiment.

Scatter Plot Interpretation:

The scatter plot displays the relationship between **host sentiment** (x-axis) and **reviewer sentiment** (y-axis).

Weak Positive Trend: There's a slight positive trend, where higher host sentiment often correlates with higher reviewer sentiment, but the relationship is not strong.

Outliers: High reviewer sentiment is observed even when host sentiment is neutral or moderately positive, implying that other factors (like amenities or property quality) influence review sentiment.

Variance in Reviewer Sentiment: The wide spread of reviewer sentiment across all host sentiment levels further supports the weak correlation.

Both the correlation matrix and scatter plot show that **host sentiment is not the primary driver** of review sentiment. While there is a positive relationship, other factors like property quality likely have a significant impact on reviews. Property managers should focus on both guest interactions and property improvements to boost review scores.

3.1. Chosen Models and Justification

Three different machine learning models were chosen for this task, each with unique strengths that make them suitable for predicting review scores:

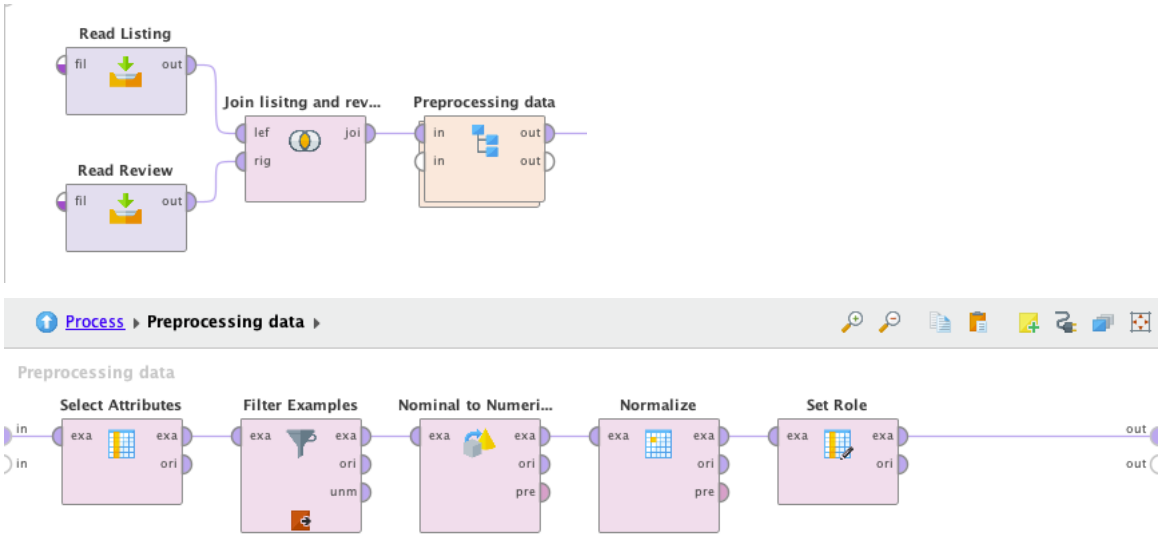
The seed I'm using is seed number: 2024

Random Forest: A robust ensemble learning model that builds multiple decision trees during training. It is well-suited for datasets with complex relationships between features. Random Forest was chosen because it handles both numerical and categorical data well, provides good generalization, and is less prone to overfitting.

Gradient Boosting: This model iteratively builds decision trees, with each new tree correcting the errors made by the previous trees. Gradient Boosting is known for its strong predictive performance, especially in regression tasks like predicting review scores. It was chosen for its ability to minimize error over multiple iterations, leading to more accurate predictions.

Decision Trees: A simpler model that splits data into branches based on decision rules. Decision Trees are easy to interpret and useful as a baseline model for comparison. They were selected to understand how a simpler approach compares to more complex ensemble methods.

These models were then combined in a **Voting Ensemble** to leverage the strengths of all three. The ensemble model was used to reduce variance and improve the overall predictive power by averaging the predictions from the individual models. Data preprocessing:



Read data: I loaded the listing and review datasets then merged them using the join operator.

Join: This operator was used to merge two datasets into one.

3.2. Data preparation (nested inside Subprocess named preprocessing data)

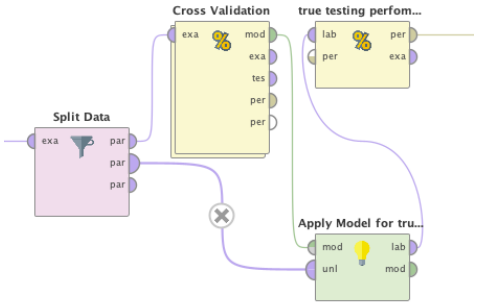
Select Attribute: This operator was used to select the attribute for the predictive model, these attributes will later be used to predict review score.

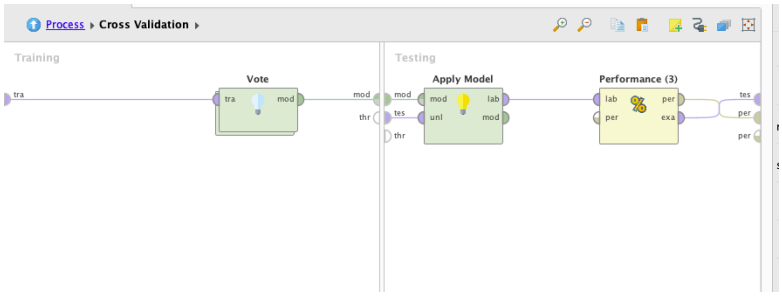
Filter Examples: This operator was used to remove missing value, clean the data for further use.

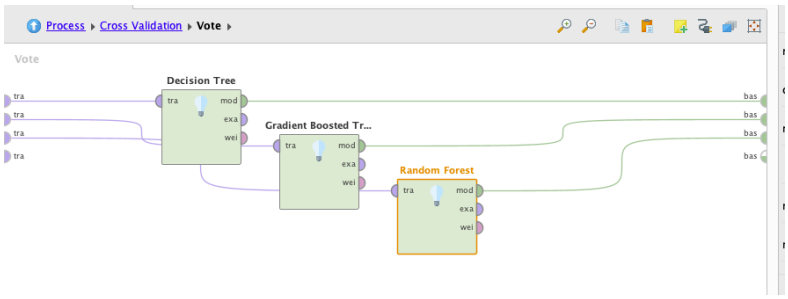
Nominal to Numerical: This was used to convert nominal (Categorical) data into numerical form, which is an important step since review score is a number, we need to convert data into number form to be able to use.

Normalize: This was used to scale the value of numerical attributes so that they fall within a specific range.

Set role: This was used to label Review score as a label for the process to understand that Review score is the target to predict.







3.3. Model training and applying model

Split data: for true testing, I have split the data into 70%/20%/10% in which 70% is for training and 20% percent is for testing both connected with Cross-validation. For the last 10%, I used to test with the true testing.

Voting Operator: this operator was created by combining the outputs of three base models: Random Forest, Gradient Boosting, and Decision Trees.

Voting Operator is nested inside Cross-validation (10-fold): this step reduces the risk of overfitting and ensures that the model generalizes well to unseen data.

Cross-Validation (10-folds): I used 10-fold cross-validation to validate the model performance on different subsets of the data. Cross-validation reduces the risk of overfitting and ensures that the model generalizes well to unseen data.

3.4. Model parameter:

Random Forest:
- Number of trees: 200.
- Maximal depth: 10.
- Apply repruning.

Gradient Boosted Trees:
- Number of trees: 100
- Maximal depth: 10

Decision Tree:
- Maximal depth: 10
- Apply repruning.

The model then feed and apply to the rest 10% of data for true testing which enhance the result and prevent overfitting.

3.5. Ensemble Voting Model

The **Voting Ensemble** model was developed using the **Voting** operator, which combines the predictions of multiple models. This method allows the final prediction to be more robust and accurate than any individual model alone.

Justification: The **Voting Ensemble** was selected because it aggregates the predictions of the Random Forest, Gradient Boosting, and Decision Tree models. Each model captures different aspects of the data, and by combining their predictions, the ensemble reduces the variance and improves the overall prediction accuracy.

Model evaluation and improvement (2 pages)

4.1. Key Metrics to consider:

- ***Root Mean Squared Error (RMSE):*** A lower RMSE indicates better overall performance.
- ***Absolute Error:*** Lower absolute error suggests prediction accuracy.
- ***Relative Error***: lower relative error values indicate a better fit of the model to the data, relative to the range of values.
- ***Relative Error Lenient:*** Like relative error but using a less strict evaluation metric

4.2. Summary of Result:

Model	***RMSE***	***Absolute Error***	***Relative Error***	***Relative error lenient***
Voting Ensemble with 3 model (Random Forest, Gradient Boosting, Decision tree)	0.046 +/- 0.000	0.033 +/- 0.032	3.74% +/- 5.62%	3.52% +/- 3.55%
Random Forest (100 Trees, 5 depths)	0.054 +/- 0.000	0.040 +/- 0.036	4.54% +/- 6.46%	4.26% +/- 3.90%
Random Forest (200 trees, 5 depths)	0.053 +/- 0.000	0.040 +/- 0.036	4.51% +/- 6.40%	4.22% +/- 3.89%
Random Forest (100 trees, 10 depths,)	0.050 +/- 0.000	0.036 +/- 0.034	4.13% +/- 5.82%	3.89% +/- 3.69%
Gradient boosted (50 trees, 5 depths)	0.055 +/- 0.000	0.041 +/- 0.036	4.64% +/- 6.47%	4.35% +/- 3.93%
Gradient Boosted (100 trees, 10 depth)	0.045 +/- 0.000	0.032 +/- 0.031	3.65% +/- 5.83%	3.45% +/- 3.46%
Decision Tree (10 depth)	0.047 +/- 0.000	0.033 +/- 0.034	3.68% +/- 6.14%	3.46% +/- 3.72%

4.3. Discussion and Justification:

Voting Ensemble Model:

The **Voting Ensemble** outperformed individual models in terms of **RMSE**, **Absolute Error**, and **Relative Error**, making it the most robust model.

Justification: By combining multiple models, the ensemble approach mitigates the weaknesses of each individual model, leading to better generalization. The low error rates suggest that the model can effectively predict the review score, balancing the complexity of Gradient Boosting and the robustness of Random Forest.

Model Tuning:

Through **hyperparameter tuning**, I experimented with different tree depths and the number of estimators for each model to find the optimal combination that minimizes the RMSE and other errors.

For example, increasing the number of trees in Random Forest to 200 reduced the RMSE from 0.054 to 0.050.

Comparison of Models:

Gradient Boosting (100 Trees, Depth 10) performed slightly better in terms of **RMSE** and **Absolute Error** compared to the ensemble but had slightly higher variance in relative error. However, its performance alone did not justify choosing it over the ensemble.

Random Forest was consistently robust, but increasing depth led to diminishing returns after 100 trees.

4.4. Which model should be used

Voting Ensemble (Random Forest, Gradient Boosted, Decision Tree):

- **The Voting Ensemble** has the best balance of metrics.
- **RMSE (0.046)**: Better than most individual models except Gradient Boosted (100 trees, depth 10).
- **Absolute Error (0.033)**: Shows competitive accuracy.

- **Relative Error (3.74%) and Lenient Relative Error (3.52%)**: These values are among the lowest, indicating good generalization.
- **Advantage**: Combines the strengths of multiple models and generally provides more stable and robust predictions.

Gradient Boosted (100 trees, depth 10):

- **RMSE (0.045)**: The lowest RMSE among all models.
- **Absolute Error (0.032)**: The lowest, indicating high accuracy.
- **Relative Error (3.65%) and Lenient Relative Error (3.46%)**: These are also among the lowest, making it a very strong contender.
- **Advantage**: Gradient Boosting tends to perform well in many cases where decision trees and random forests might struggle.

The **Voting Ensemble** model performed the best across all key metrics, demonstrating a lower RMSE and absolute error compared to the individual models. The ensemble approach combined the strengths of each model, resulting in better generalization and more accurate predictions. The Random Forest model, while robust, performed slightly worse than the ensemble, likely due to its reliance on a single type of learner. Gradient Boosting produced comparable results but had a higher computational cost.

4.5. Performance Improvements

Several steps were taken to improve model performance:

Feature Selection: Only the most relevant features were included, reducing noise and improving accuracy.

Cross-Validation: The use of 5-fold cross-validation helped avoid overfitting and ensured that the model generalizes well to new data.

Parameter Tuning: By adjusting key parameters, such as the number of trees and tree depth, the model was optimized for both accuracy and generalization.

4.6. Impact of Predictive Modelling on Decision-Making

The predictive model provides key insights for property managers, helping them understand how different factors contribute to review scores.

Price: The model can highlight the optimal price range that maximizes review scores, helping property managers adjust pricing to attract guests while maintaining high satisfaction levels.

Property Type and Room Type: By understanding how different property and room types of influence reviews, managers can cater their offerings to specific guest preferences, potentially improving review scores.

Host Sentiment: The sentiment analysis component of the model enables hosts to gauge how their interactions impact guest reviews and make improvements where necessary.

Overall, the predictive model empowers property managers to make data-driven decisions, resulting in enhanced guest experiences and higher review scores, which ultimately lead to better property performance and increased revenue.